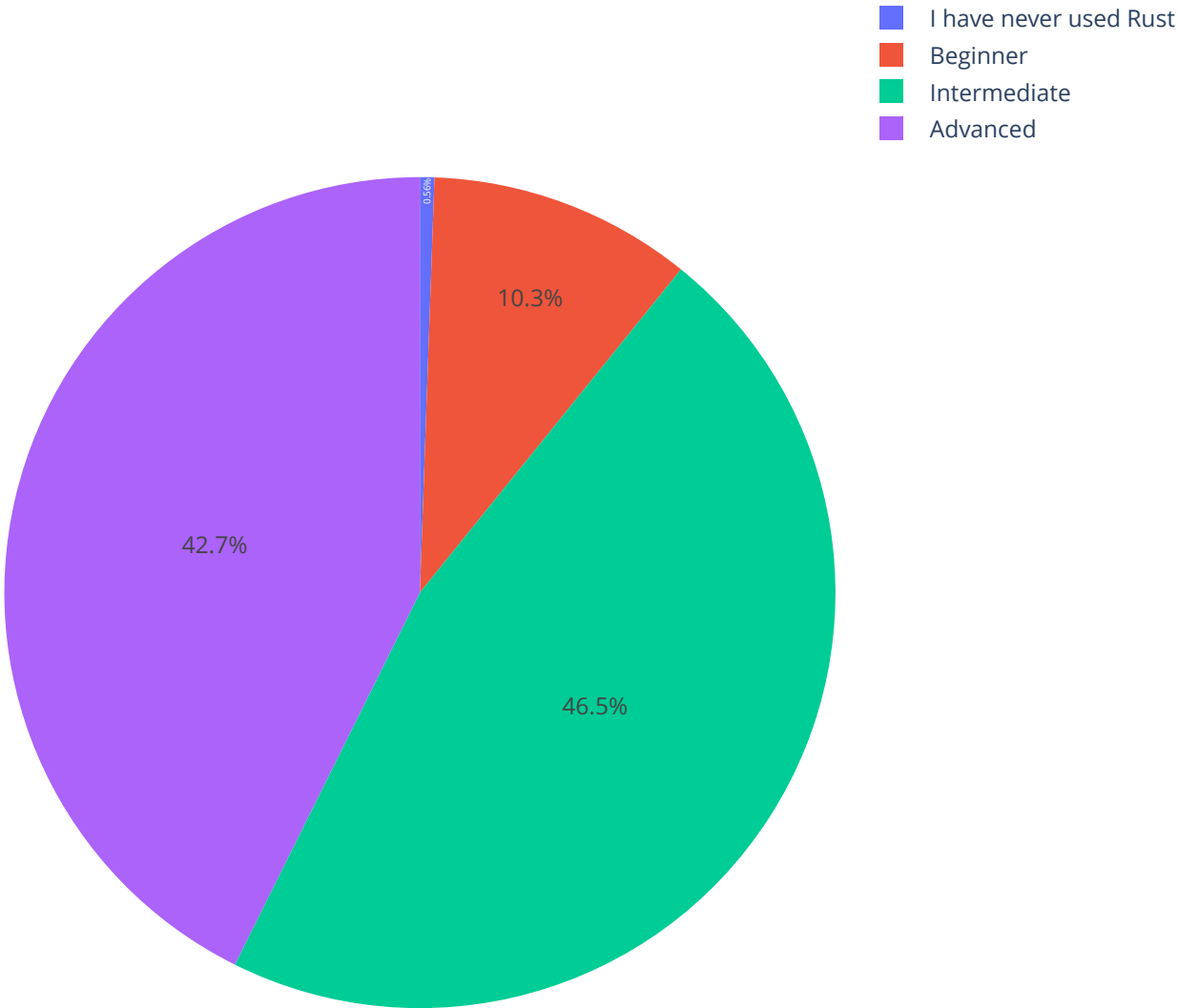


Rust Debugging survey 2026 report

How would you rate your Rust expertise?

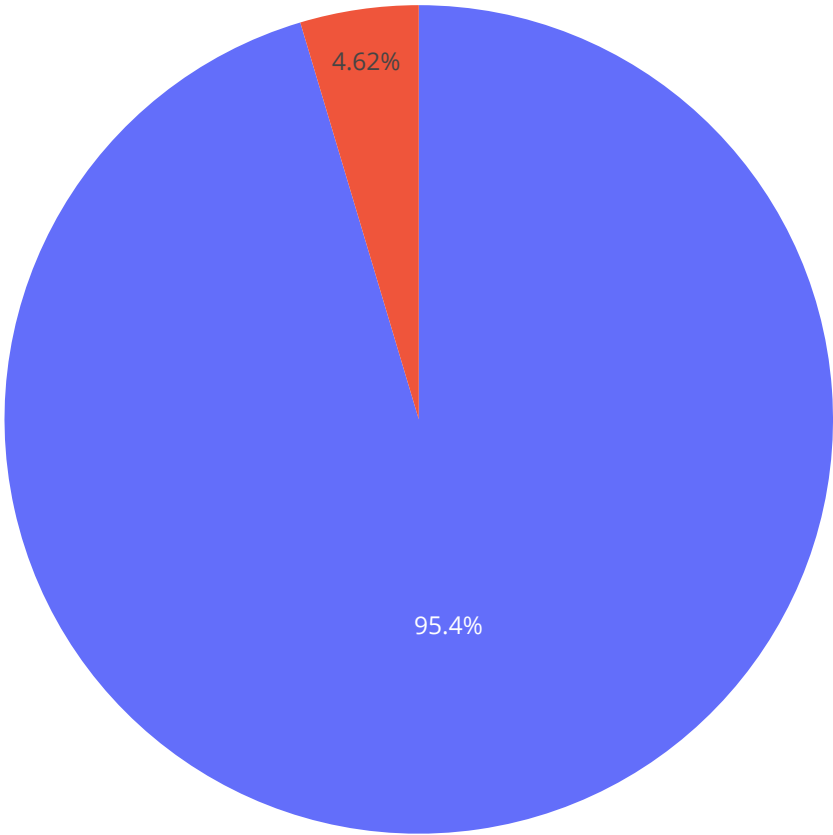
(total responses = 2321)



Do you currently use Rust?

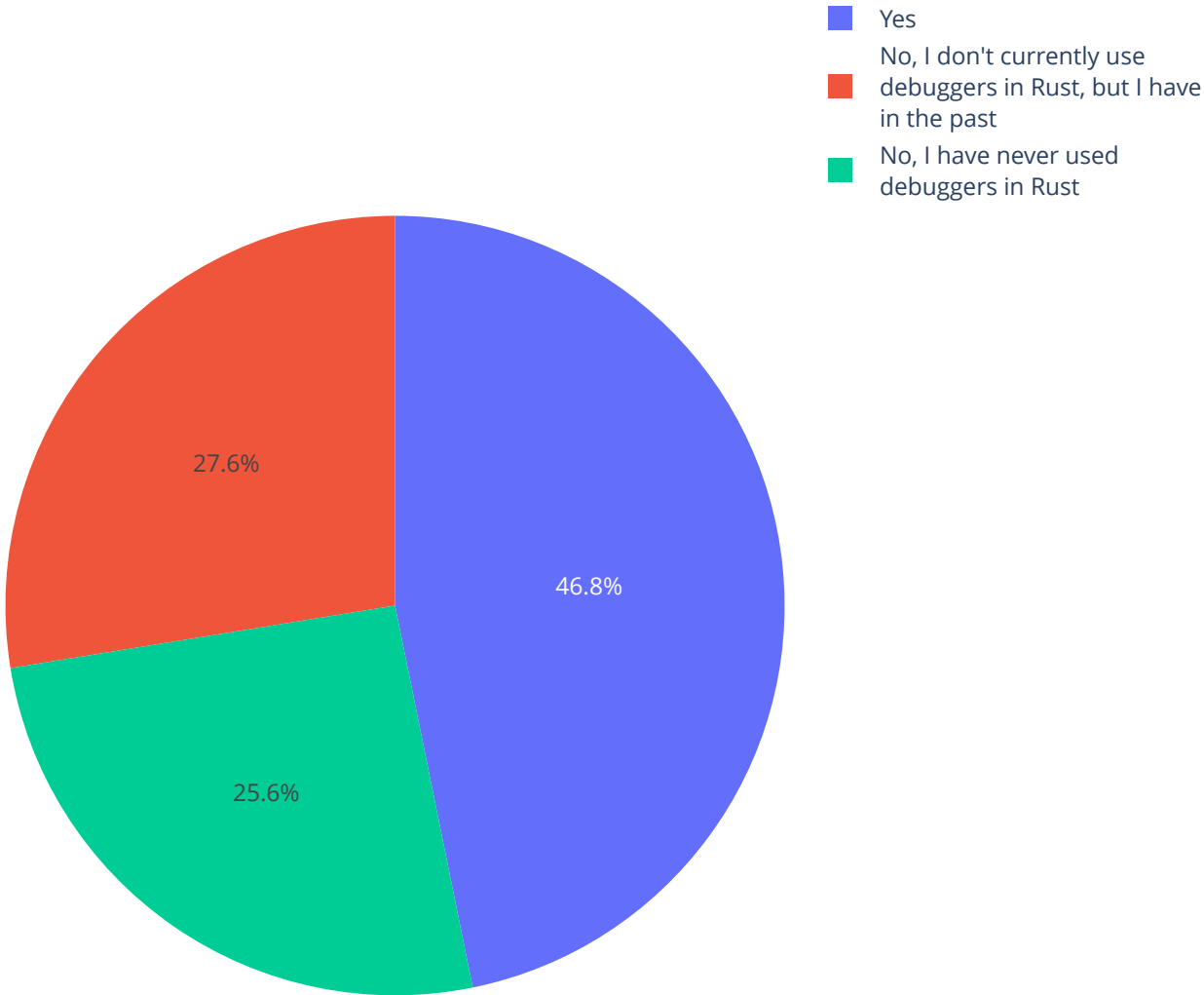
(total responses = 2295)

- Yes, I use Rust
- No, I don't currently use Rust, but I have in the past



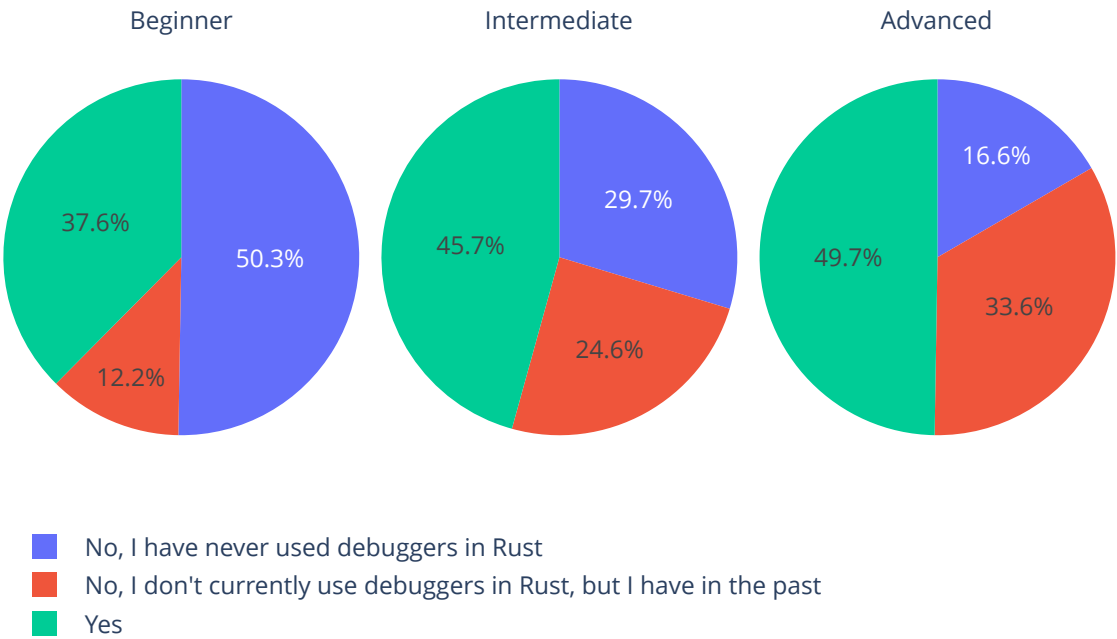
Do you use debuggers in Rust?

(total responses = 2173)



Do you use debuggers in Rust? (based on expertise)

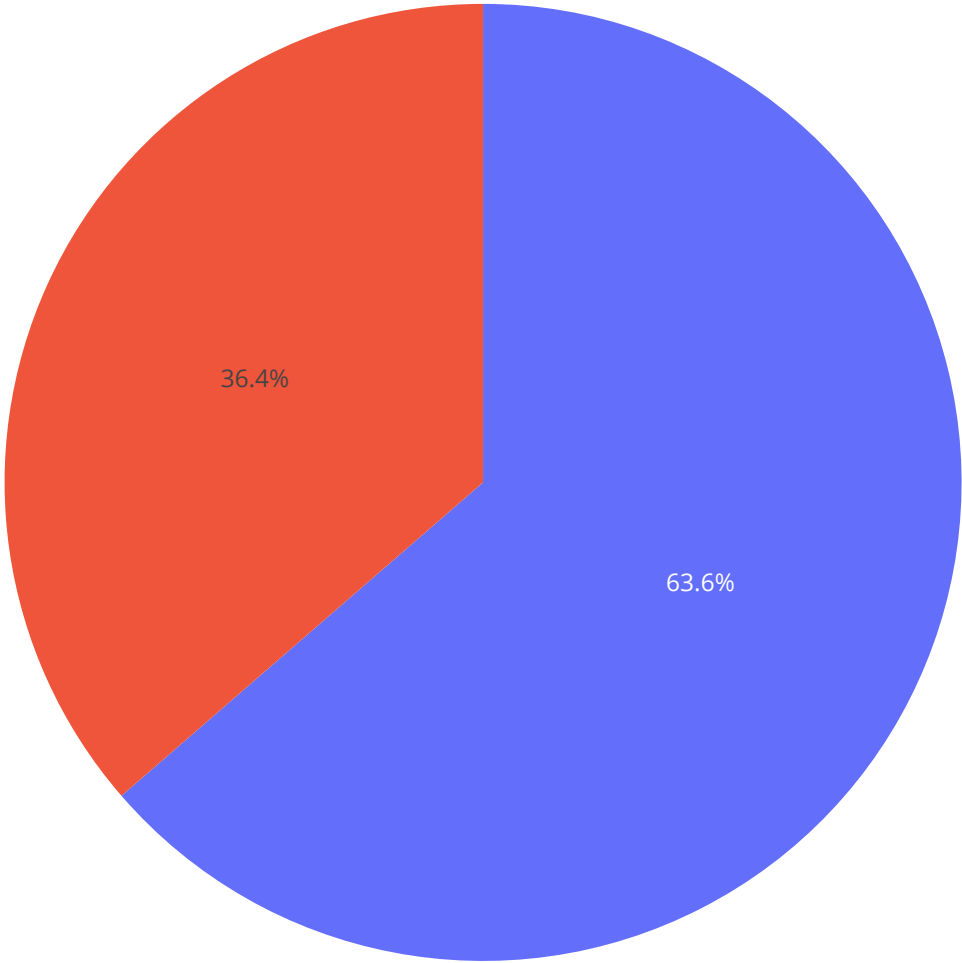
(total responses = 2173)



Did you use debuggers in Rust?

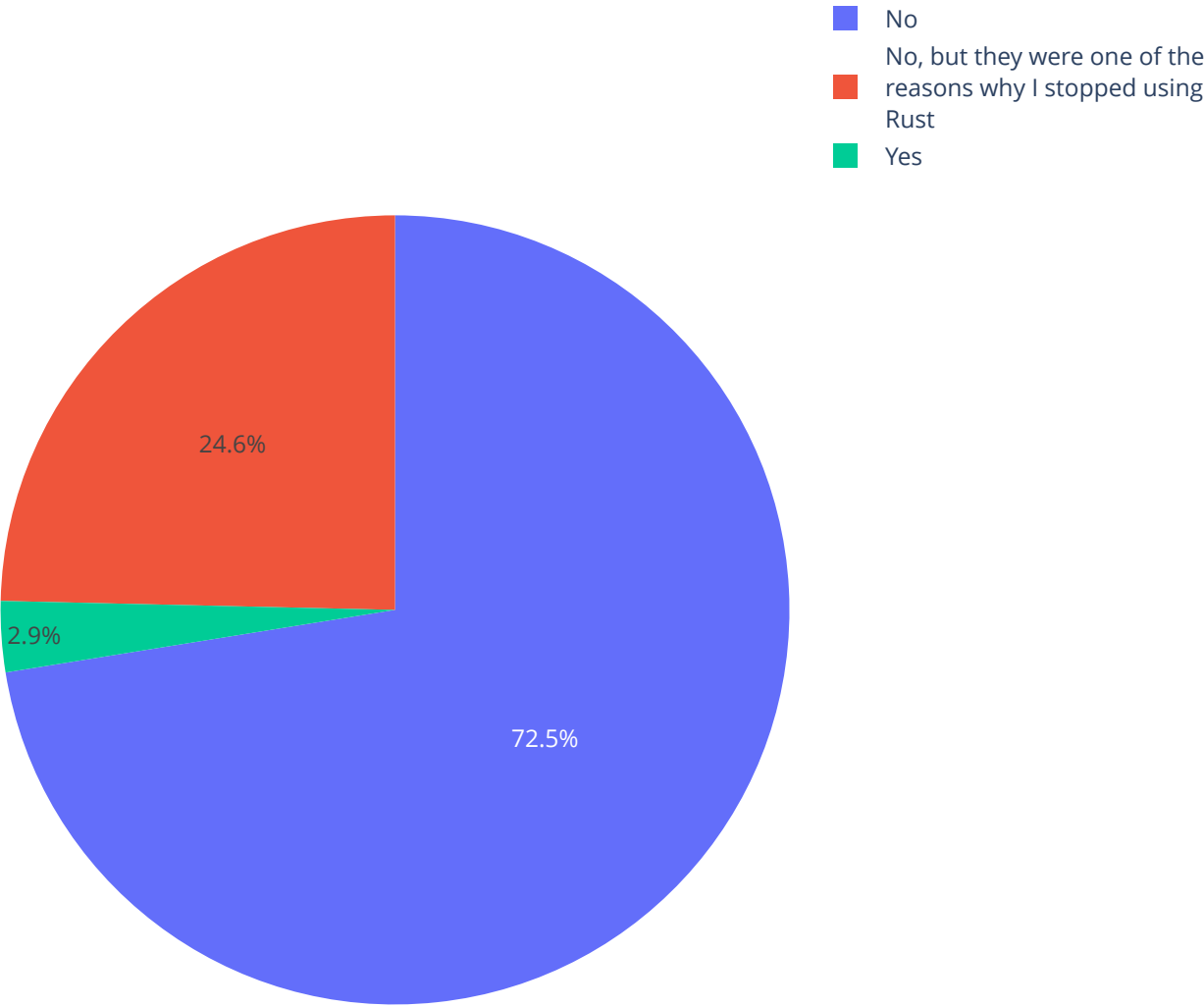
(total responses = 110)

- Yes, in the past
- No, never



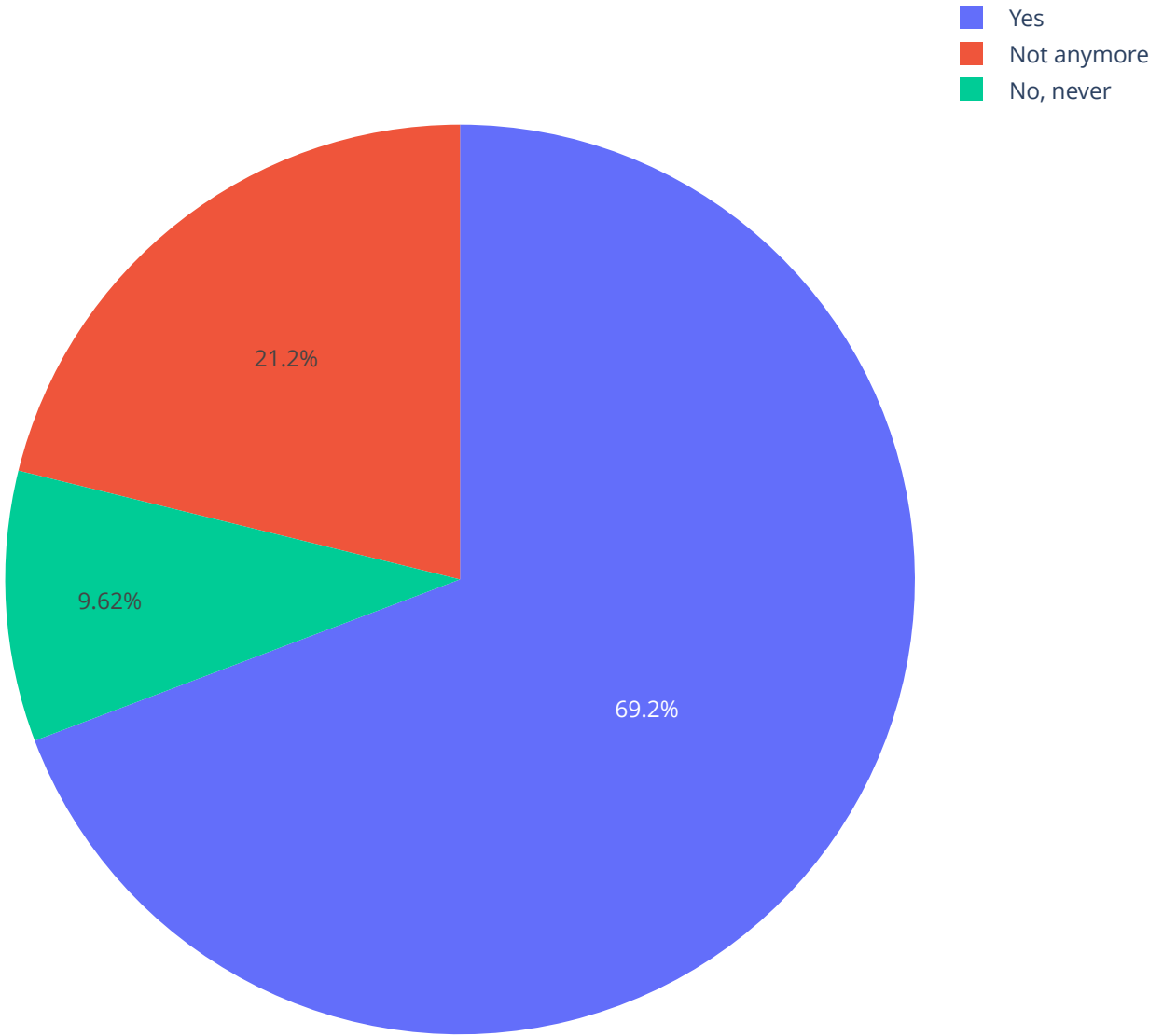
Were issues with debugging support the primary reason why you stopped using Rust?

(total responses = 69)



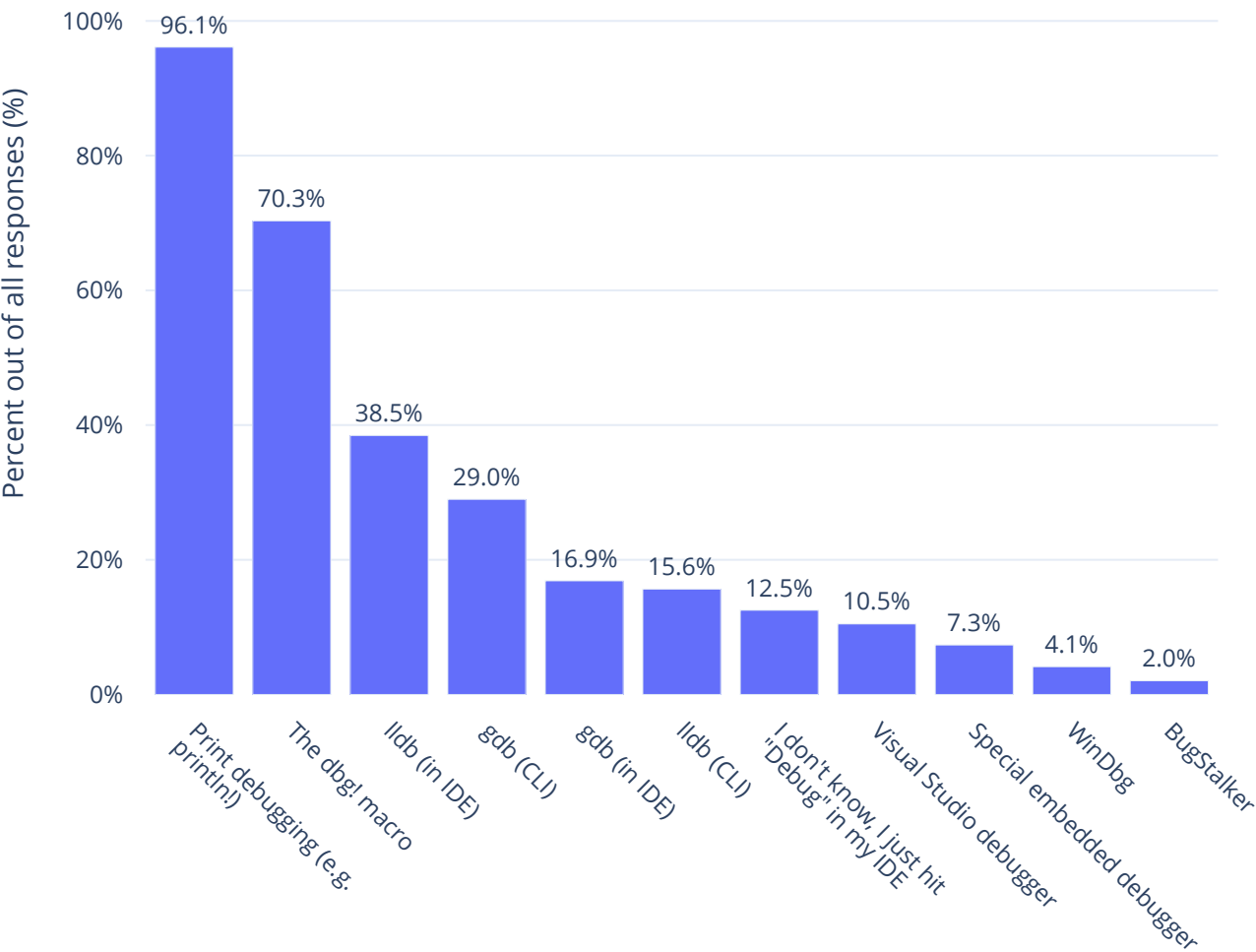
Do you use debuggers in other programming languages?

(total responses = 52)



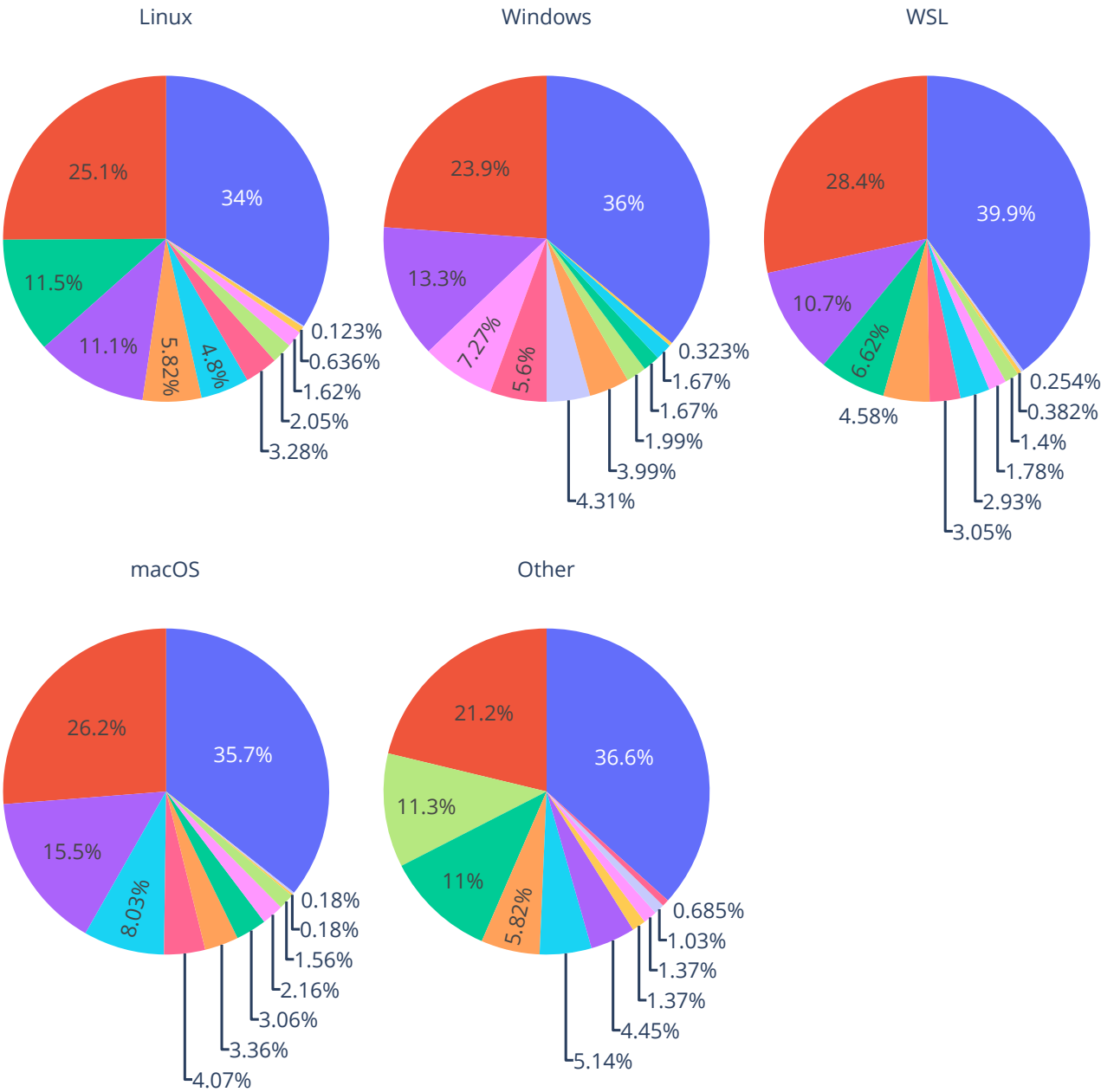
What tools and workflows do you use to debug Rust programs?

(total responses = 2109, multiple answers)



What tools and workflows do you use to debug Rust programs on which operating systems?

(total responses = 2109)



- Print debugging (e.g. println!)
- gdb (CLI)
- gdb (in IDE)
- I don't know, I just hit "Debug" in my IDE
- Visual Studio debugger
- WinDbg
- The dbg! macro
- lldb (in IDE)
- lldb (CLI)
- Special embedded debugger
- BugStalker

What tools and workflows do you use to debug Rust programs on which operating systems?

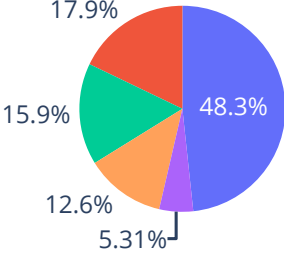
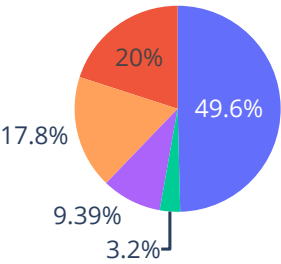
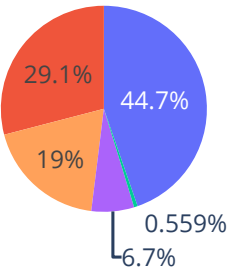
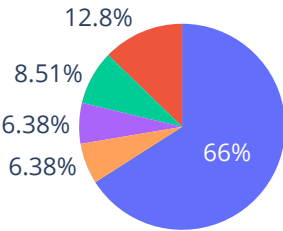
(total responses = 2109)

BugStalker

I don't know

Print debugging

Special embedded debugger

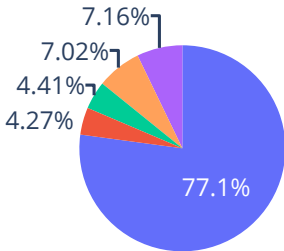
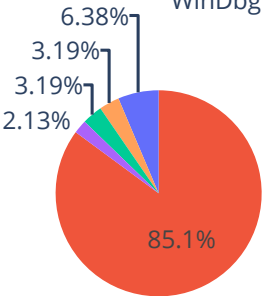
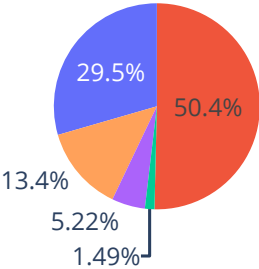
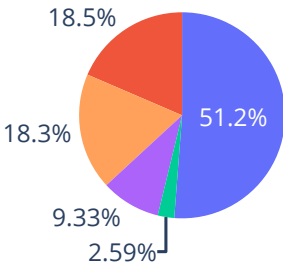


The dbg! macro

Visual Studio debugger

WinDbg

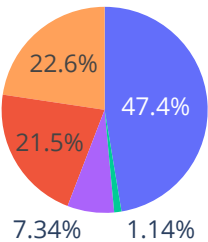
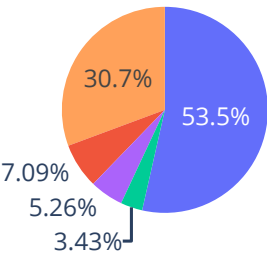
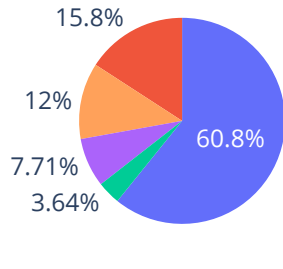
gdb (CLI)



gdb (in IDE)

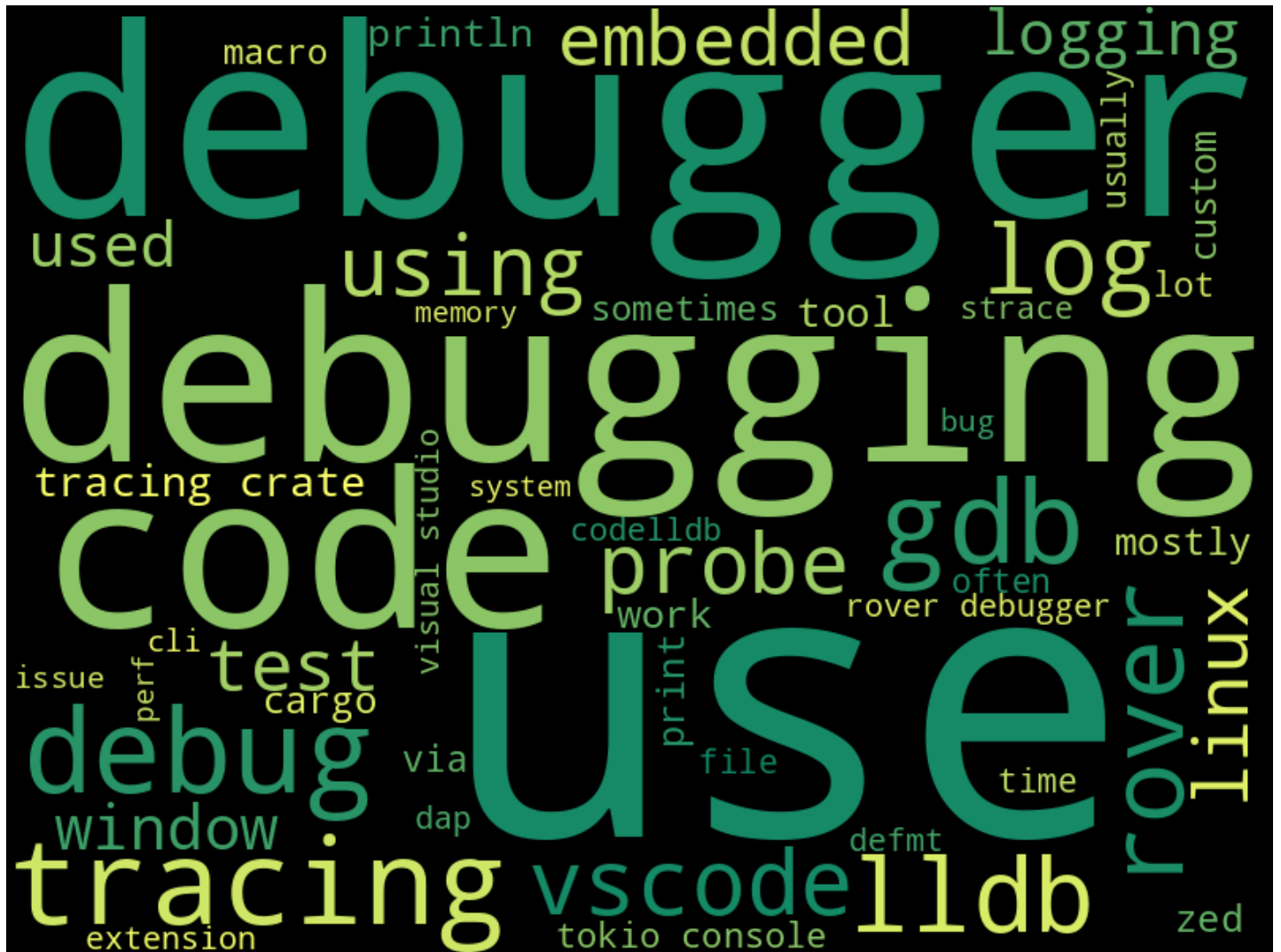
lldb (CLI)

lldb (in IDE)



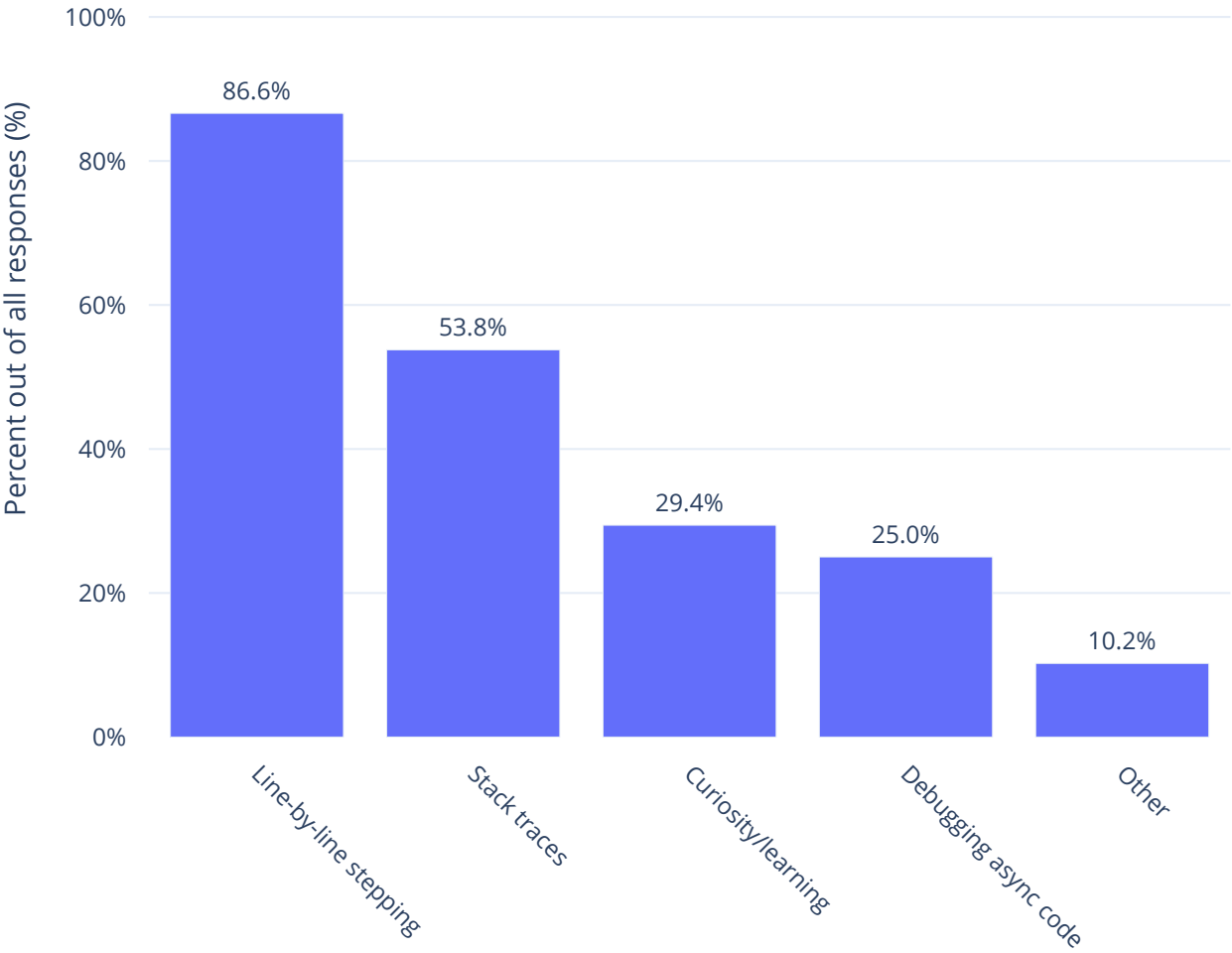
Linux Windows Other WSL macOS

Wordcloud of open answers for the previous chart:



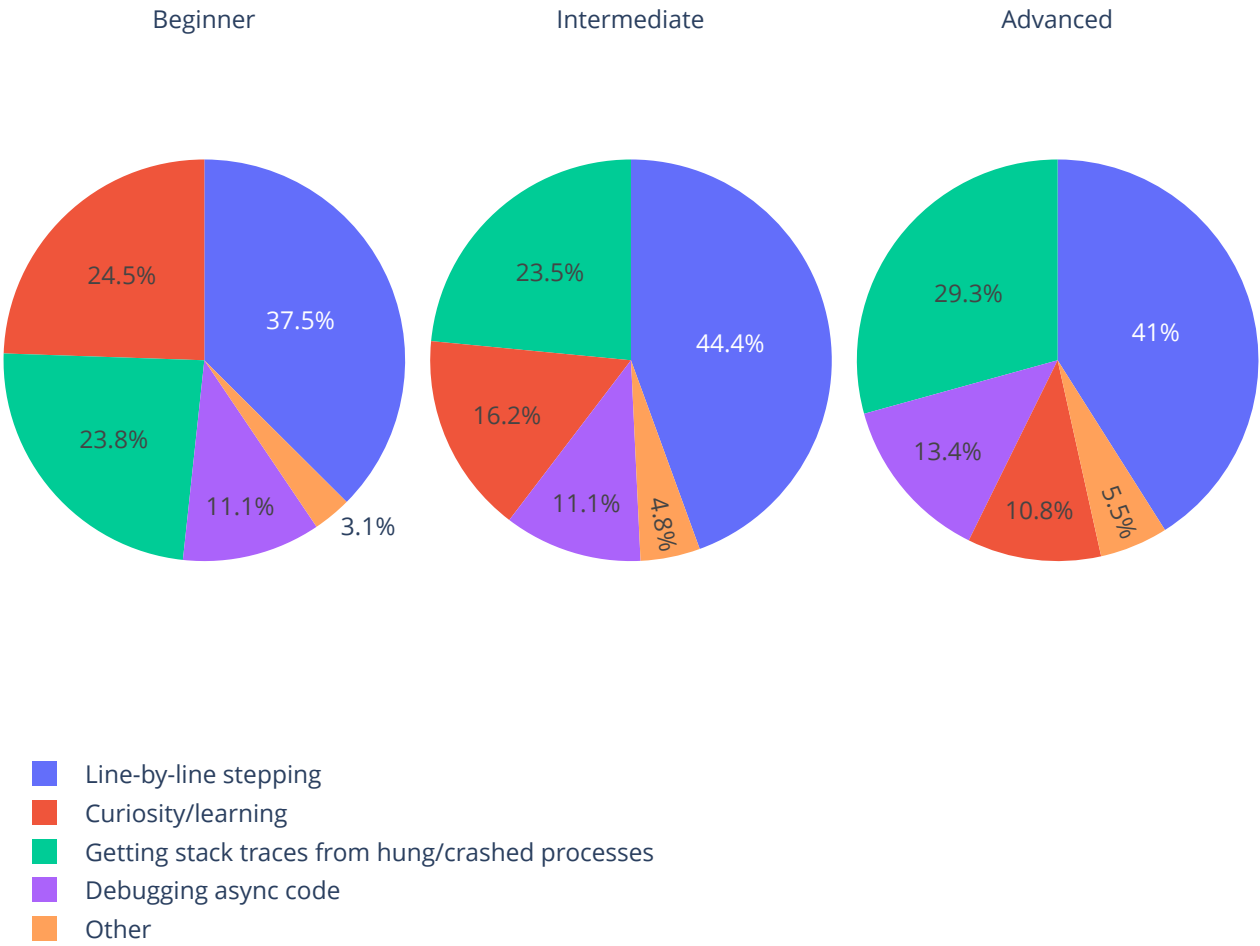
What are you using debuggers for?

(total responses = 1860, multiple answers)



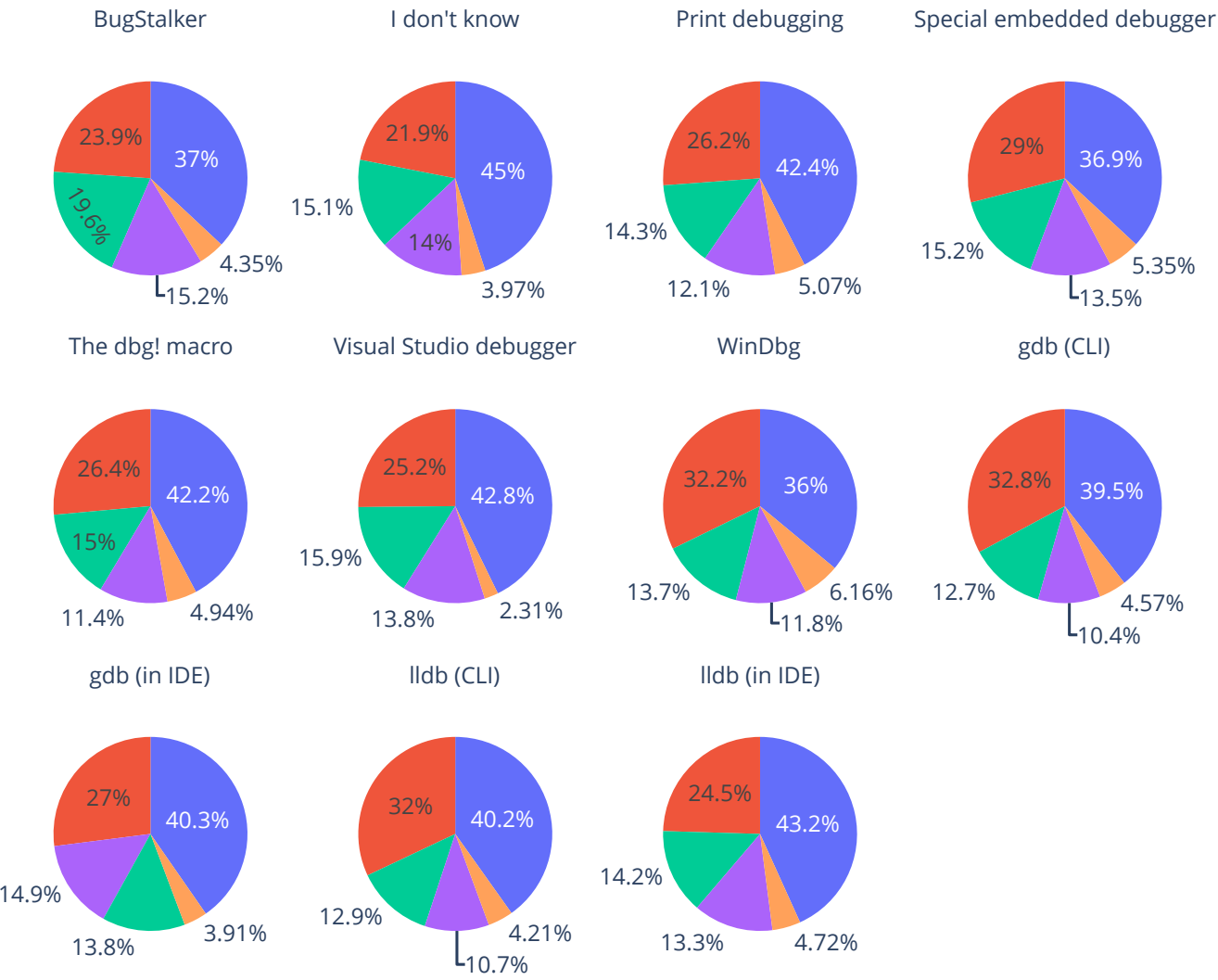
What are you using debuggers for? (based on expertise)

(total responses = 1860)



What are you using debuggers for? (based on debugger)

(total responses = 1860)

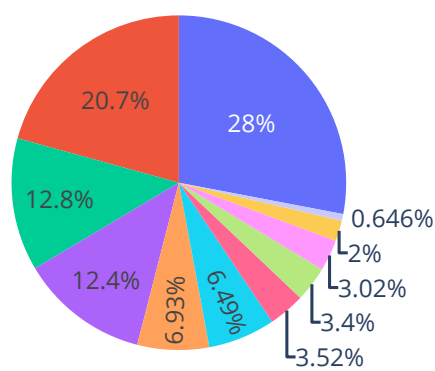


- Line-by-line stepping
- Getting stack traces from hung/crashed processes
- Curiosity/learning
- Debugging async code
- Other

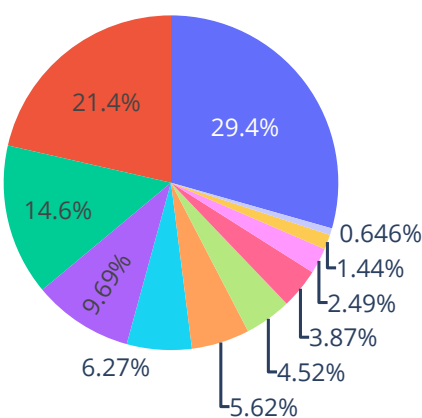
What are you using debuggers for? (based on debugger)

(total responses = 1860)

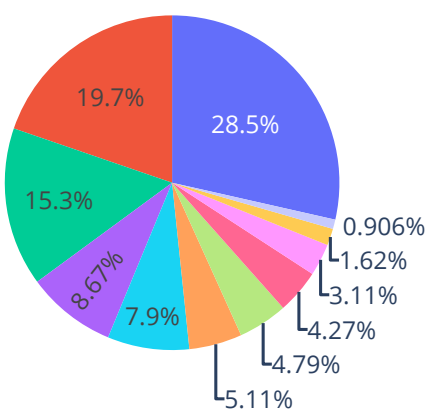
Stack traces



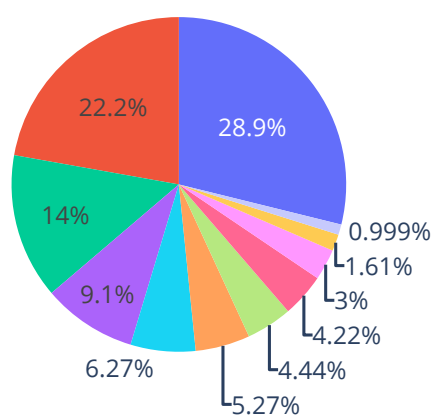
Line-by-line stepping



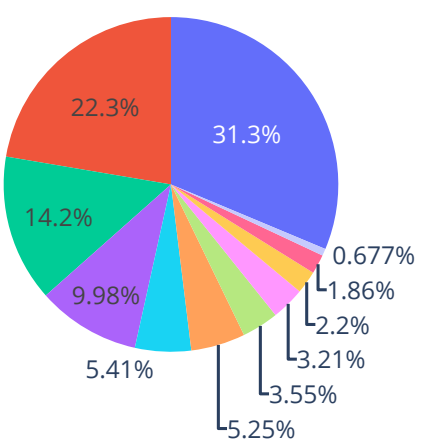
Debugging async code



Curiosity/learning

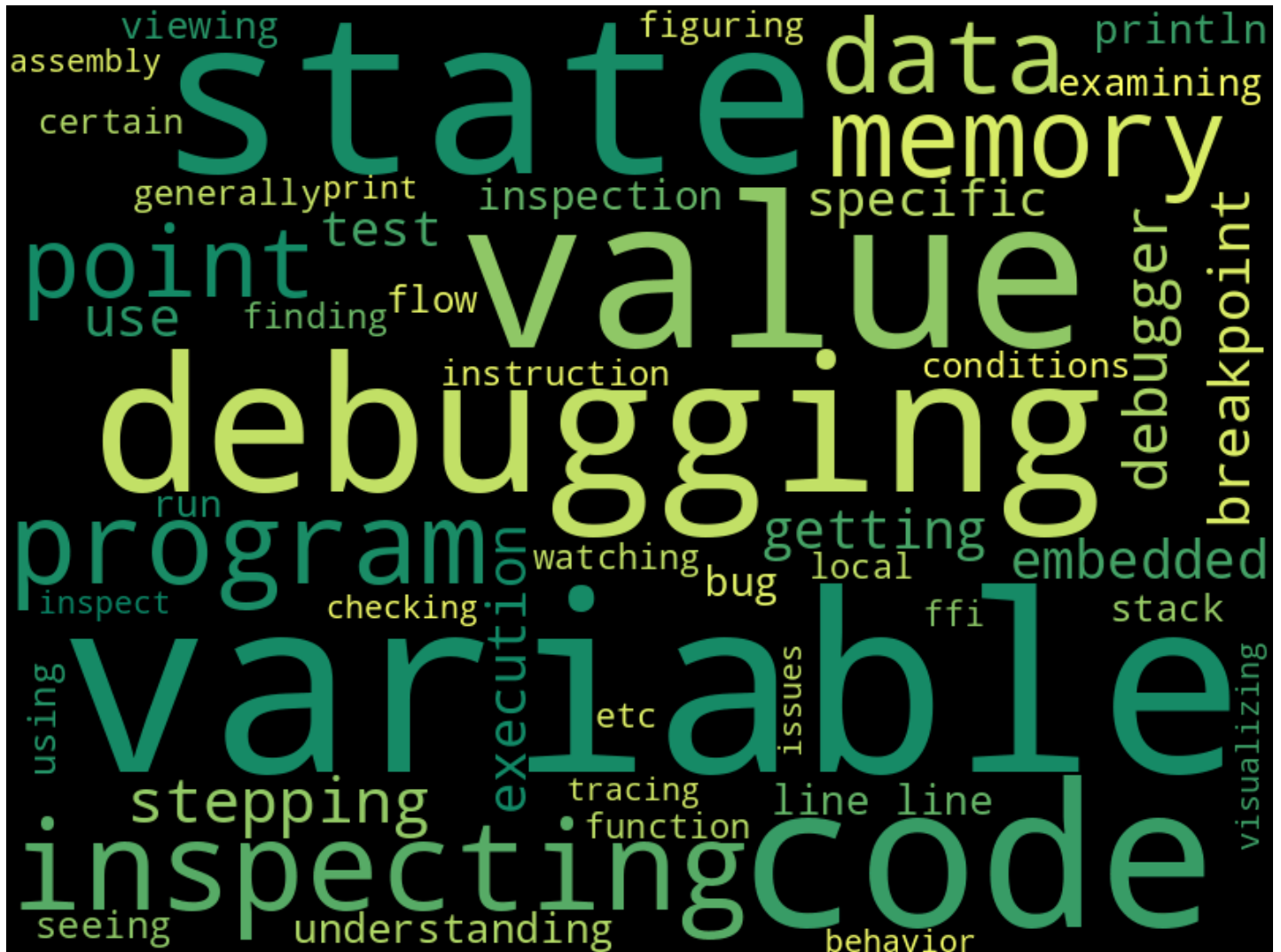


Other



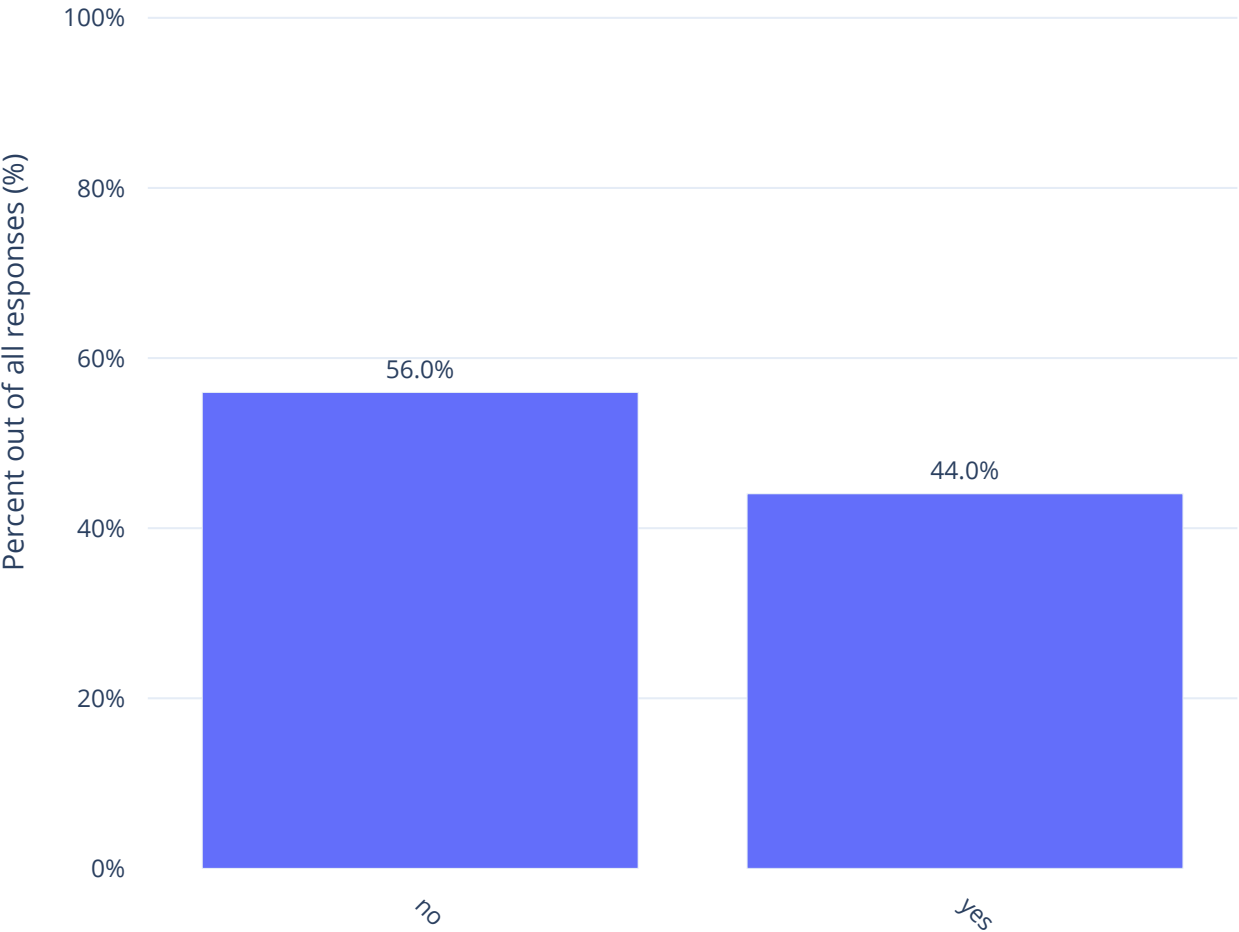
- Print debugging
- The dbg! macro
- lldb (in IDE)
- gdb (CLI)
- lldb (CLI)
- gdb (in IDE)
- Visual Studio debugger
- I don't know
- Special embedded debugger
- WinDbg
- BugStalker

Wordcloud of open answers for the previous chart:



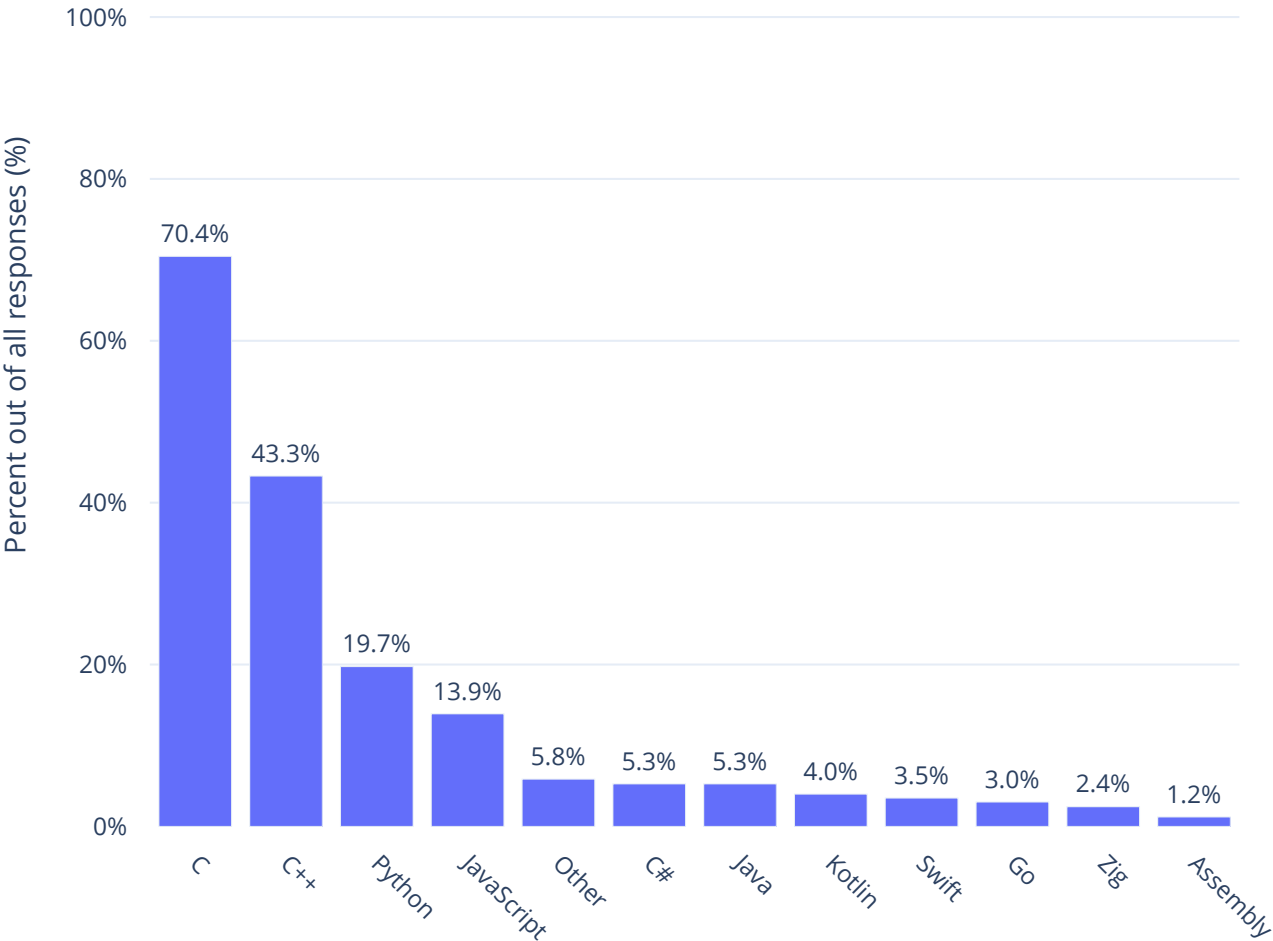
Do you debug programs that combine Rust with other languages?

(total responses = 2334, single answer)



If you do, with which languages?

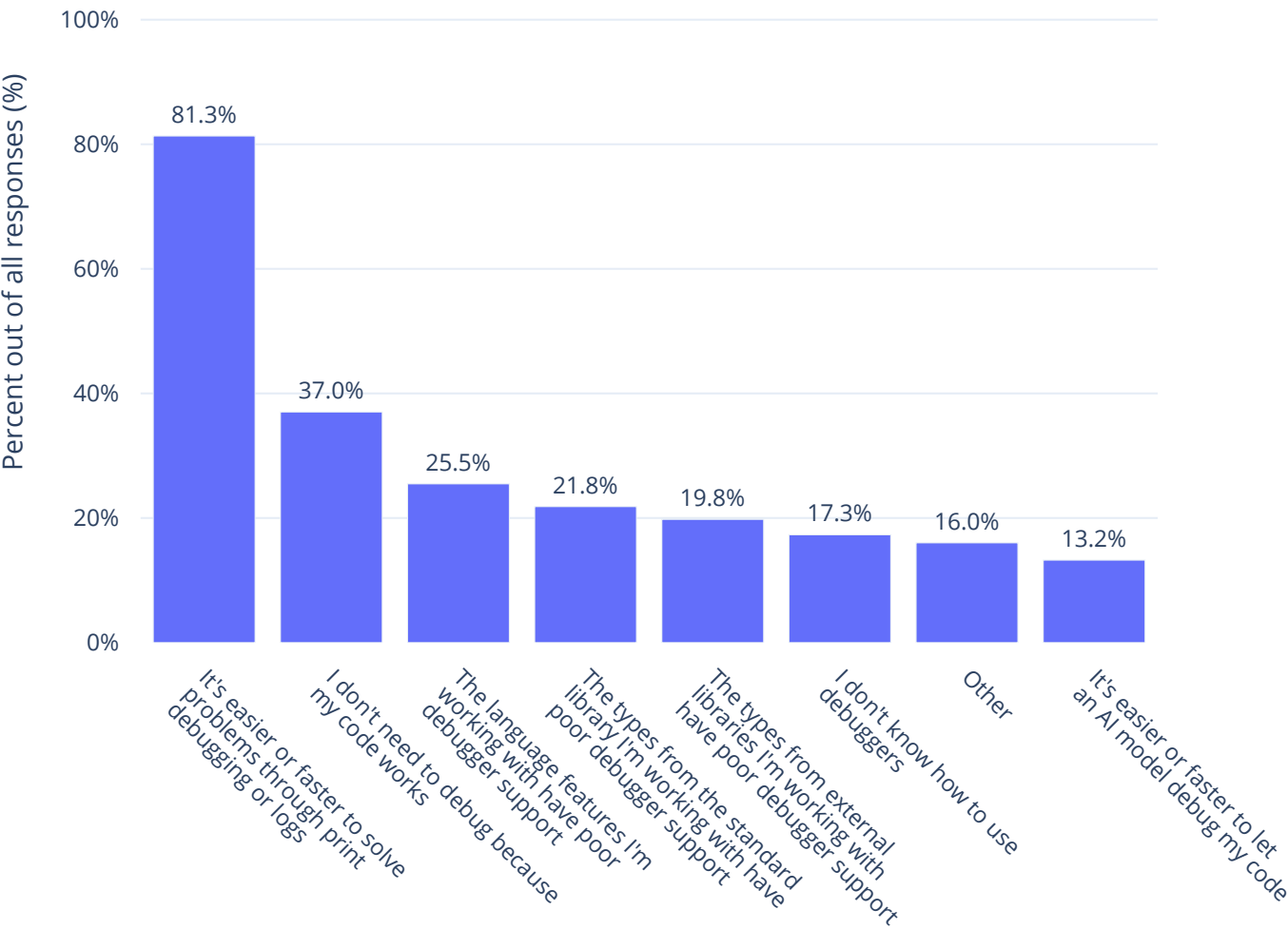
(total responses = 1028, multiple answers)



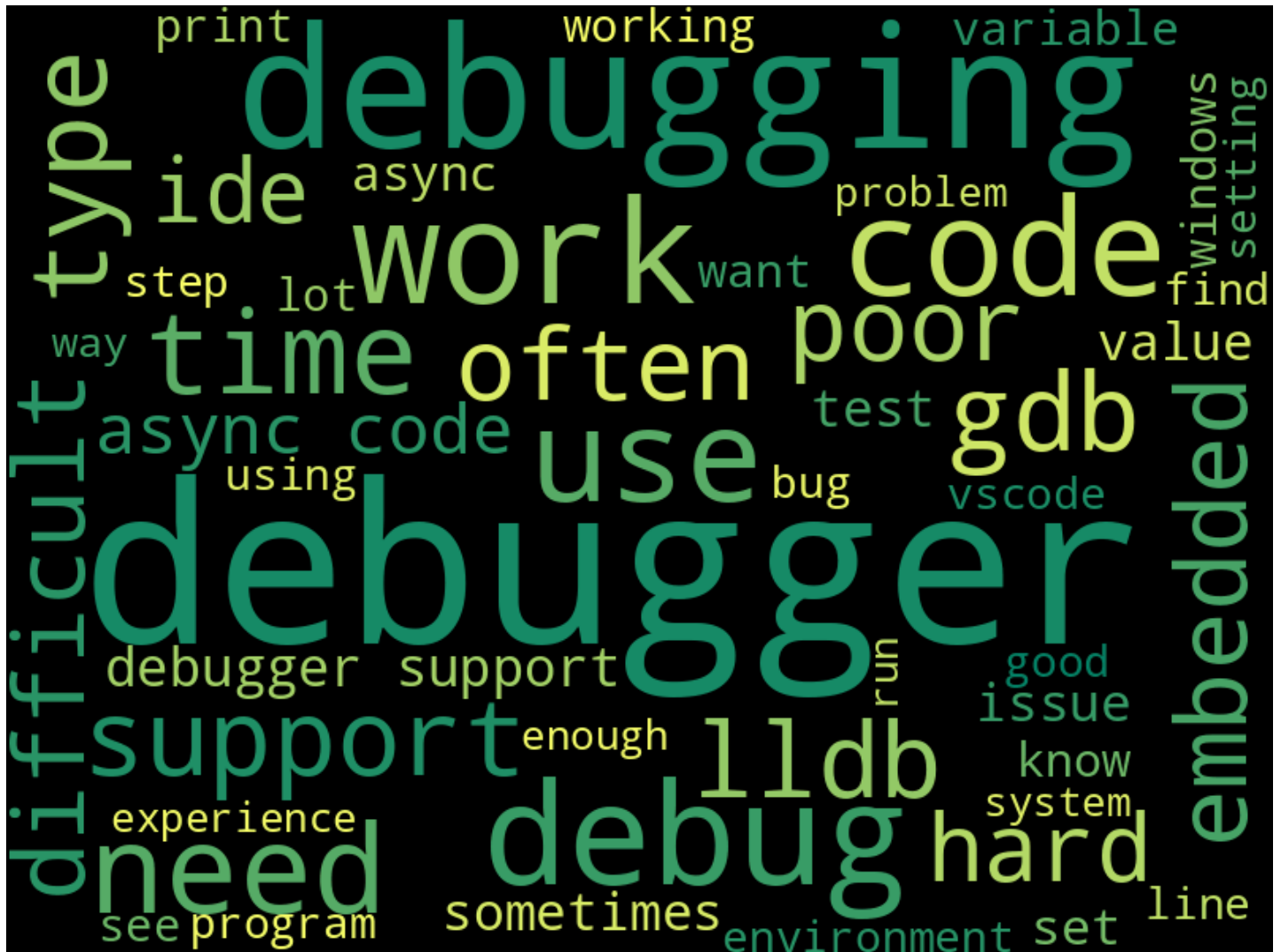
Wordcloud of open answers for the previous chart:

When you DON'T use a debugger, why don't you?

(total responses = 1943, multiple answers)



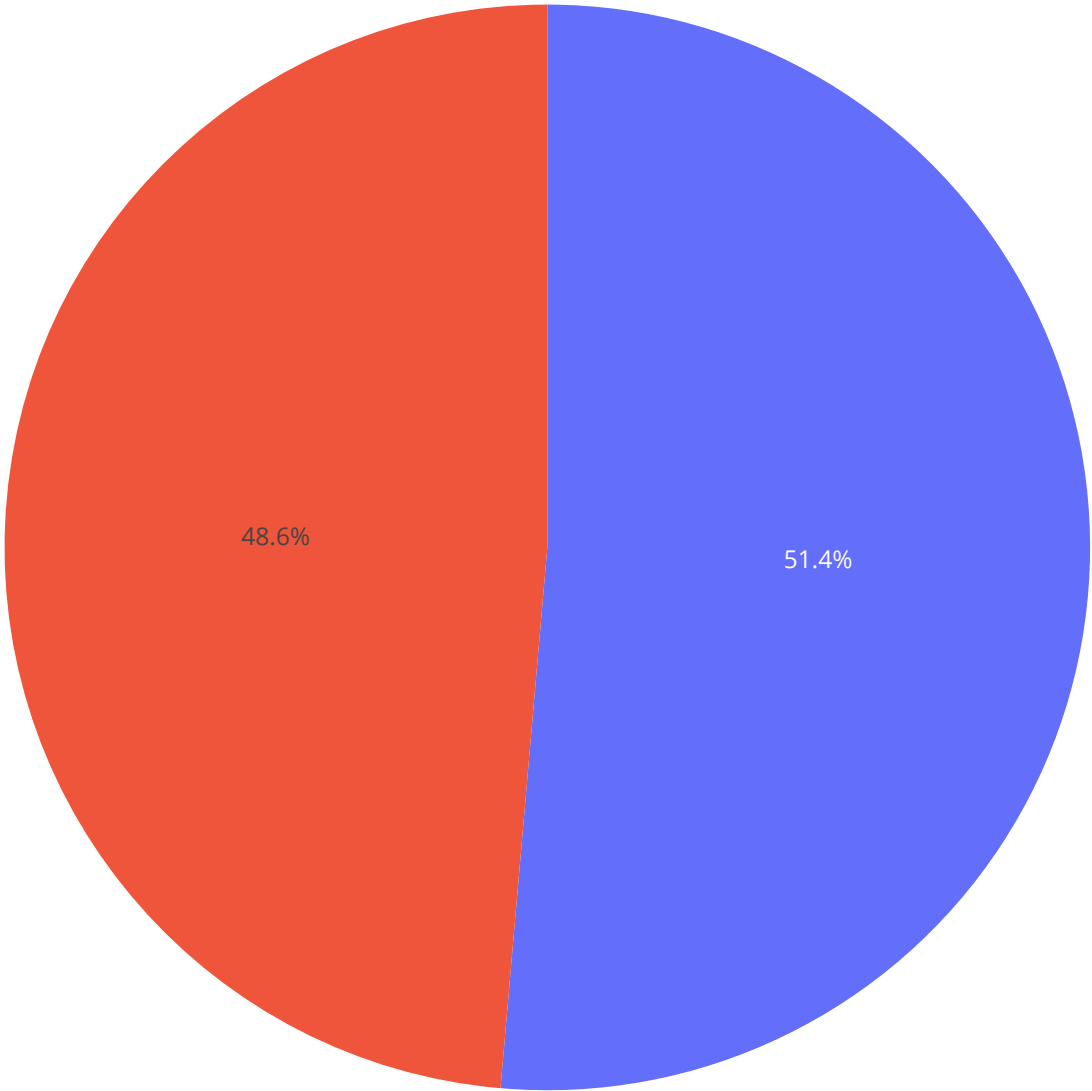
Wordcloud of open answers for the previous chart:



Do you experience any issues when trying to step through code with your debugger?

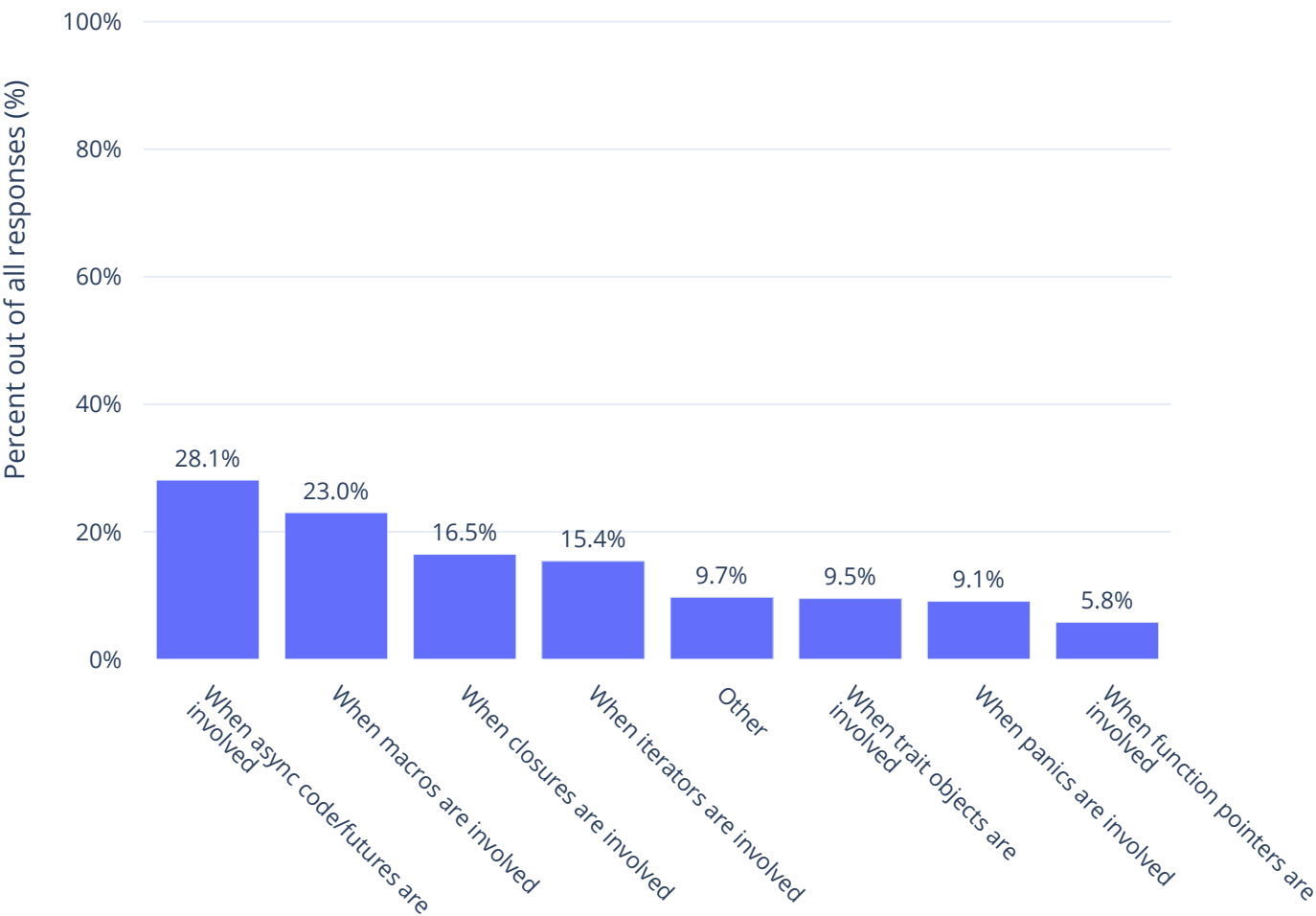
(total responses = 2036)

- Yes
- No

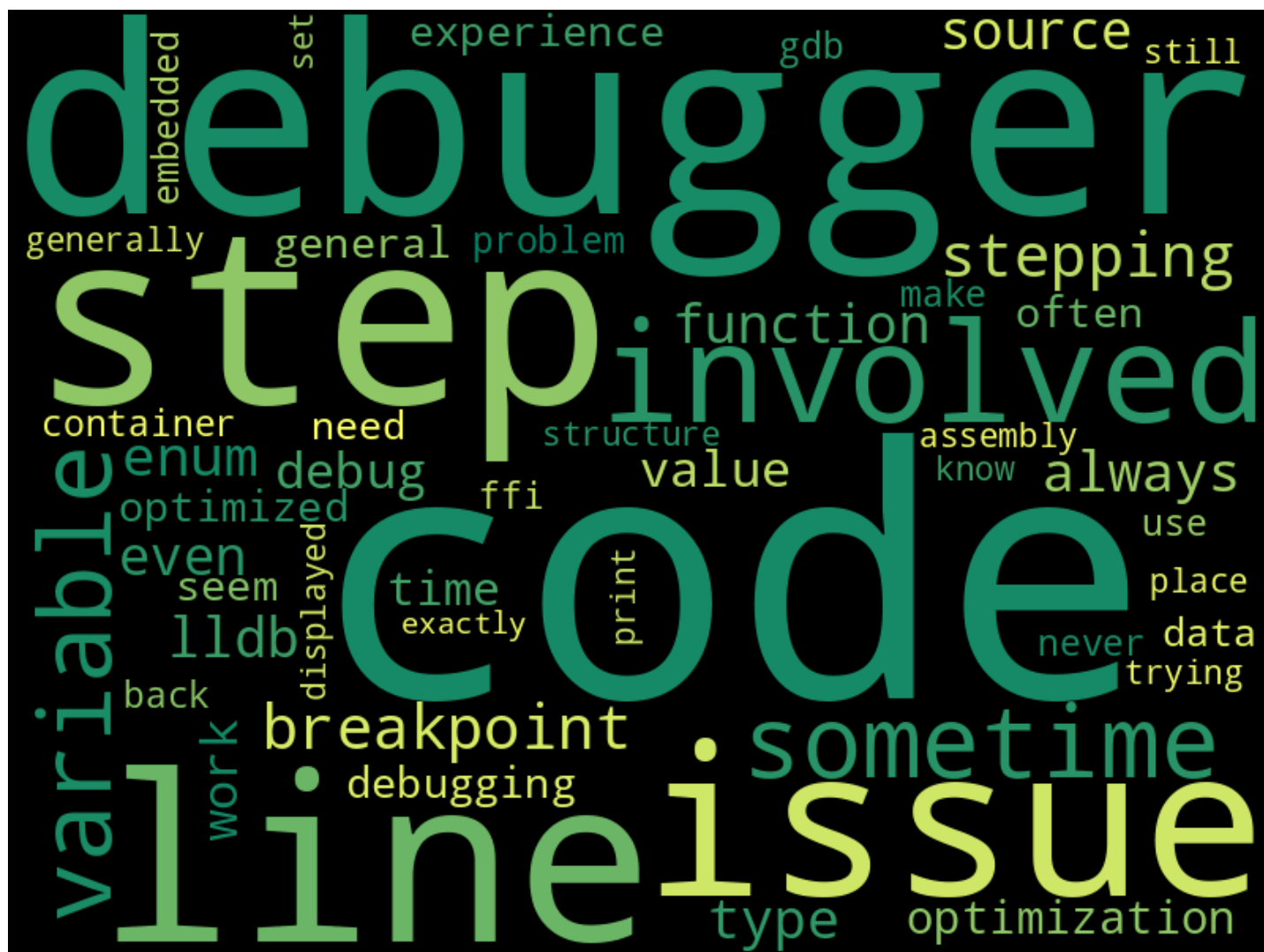


When do you experience issues with trying to step through code with your debugger?

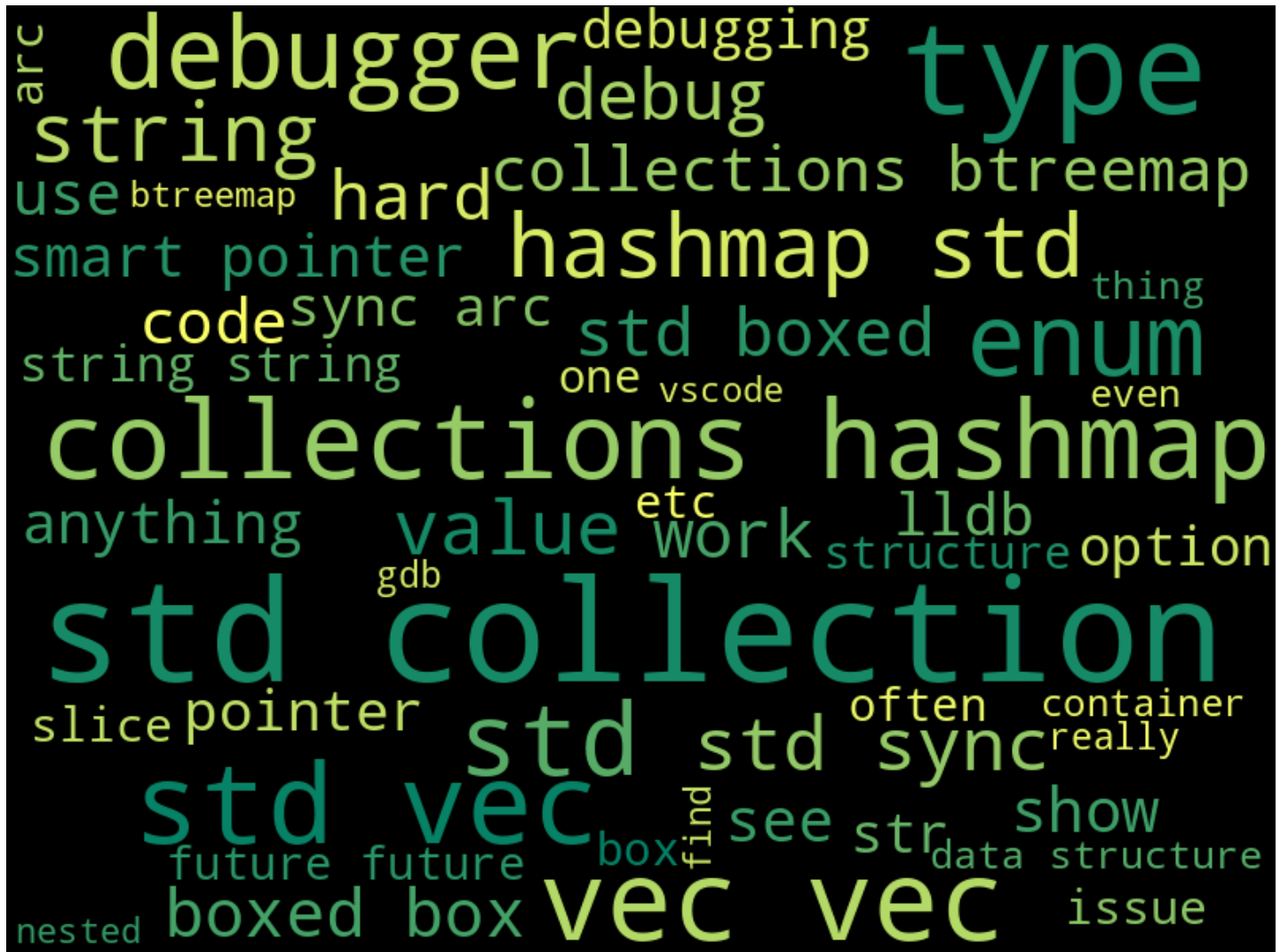
(total responses = 2036, multiple answers)



Wordcloud of open answers for the previous chart:

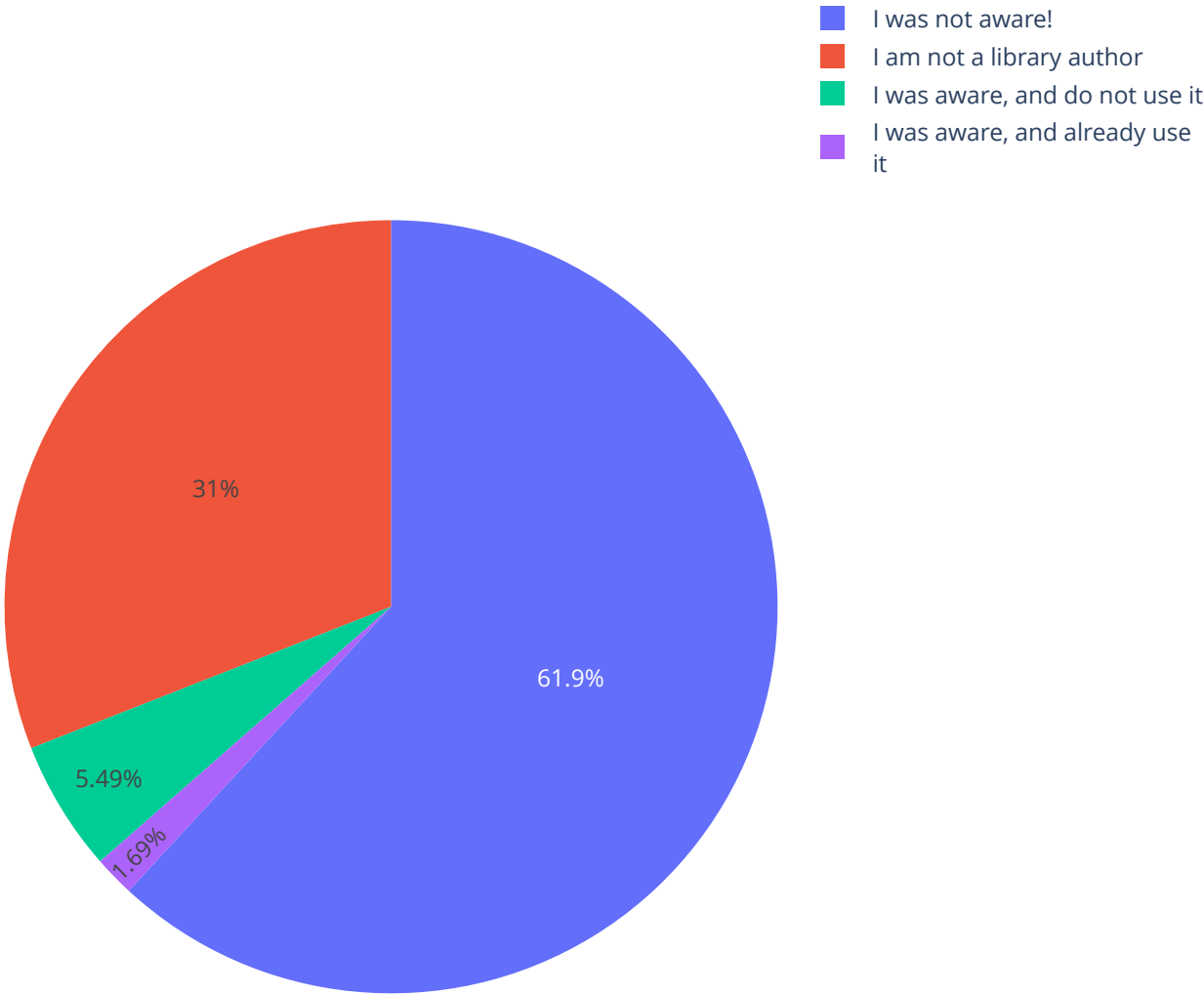


Wordcloud of open answers for the previous chart:



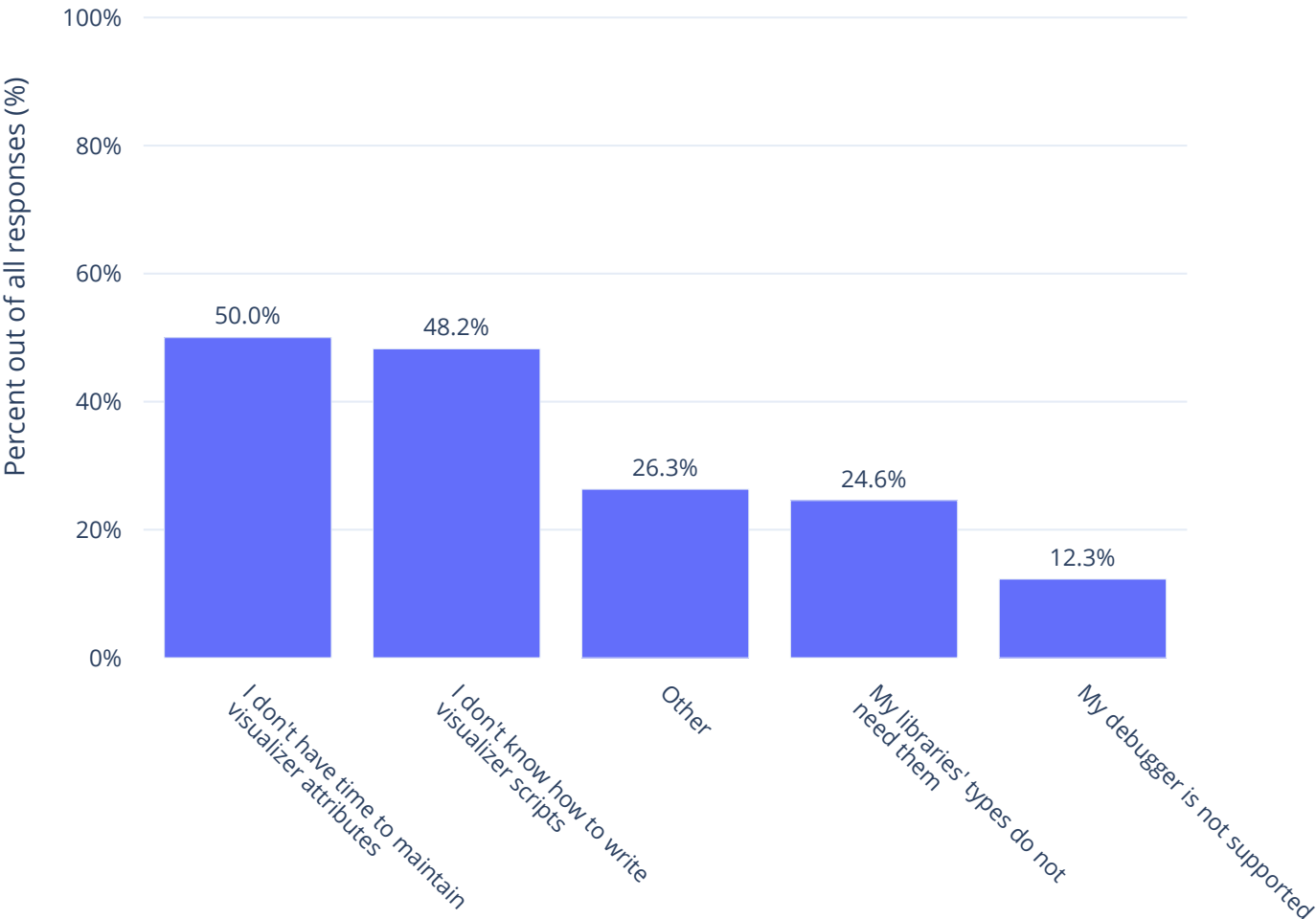
If you are a library author, are you aware of and using the debugger visualizer attribute?

(total responses = 1896)

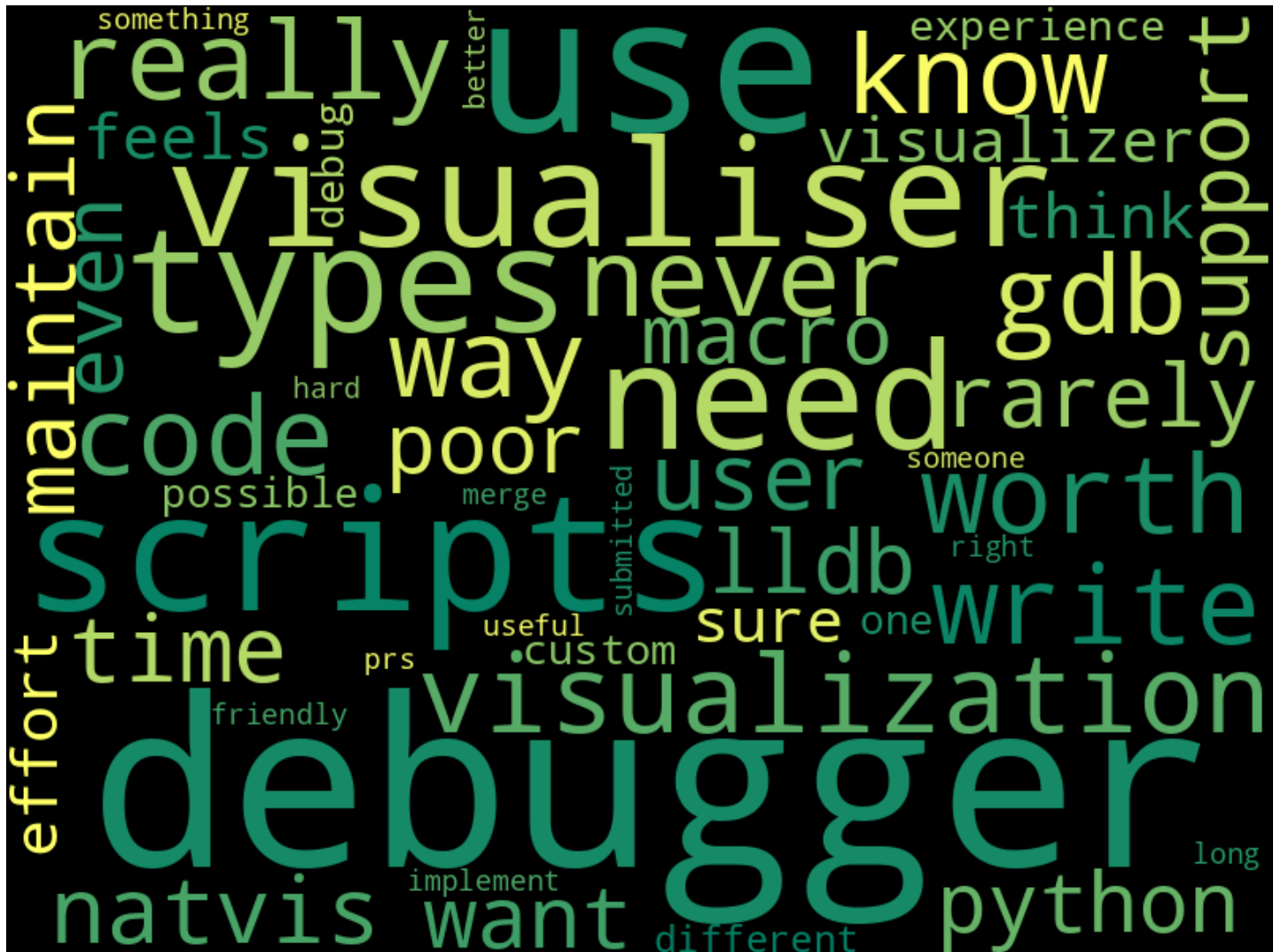


Why don't you use the debugger visualizer attribute?

(total responses = 114, multiple answers)

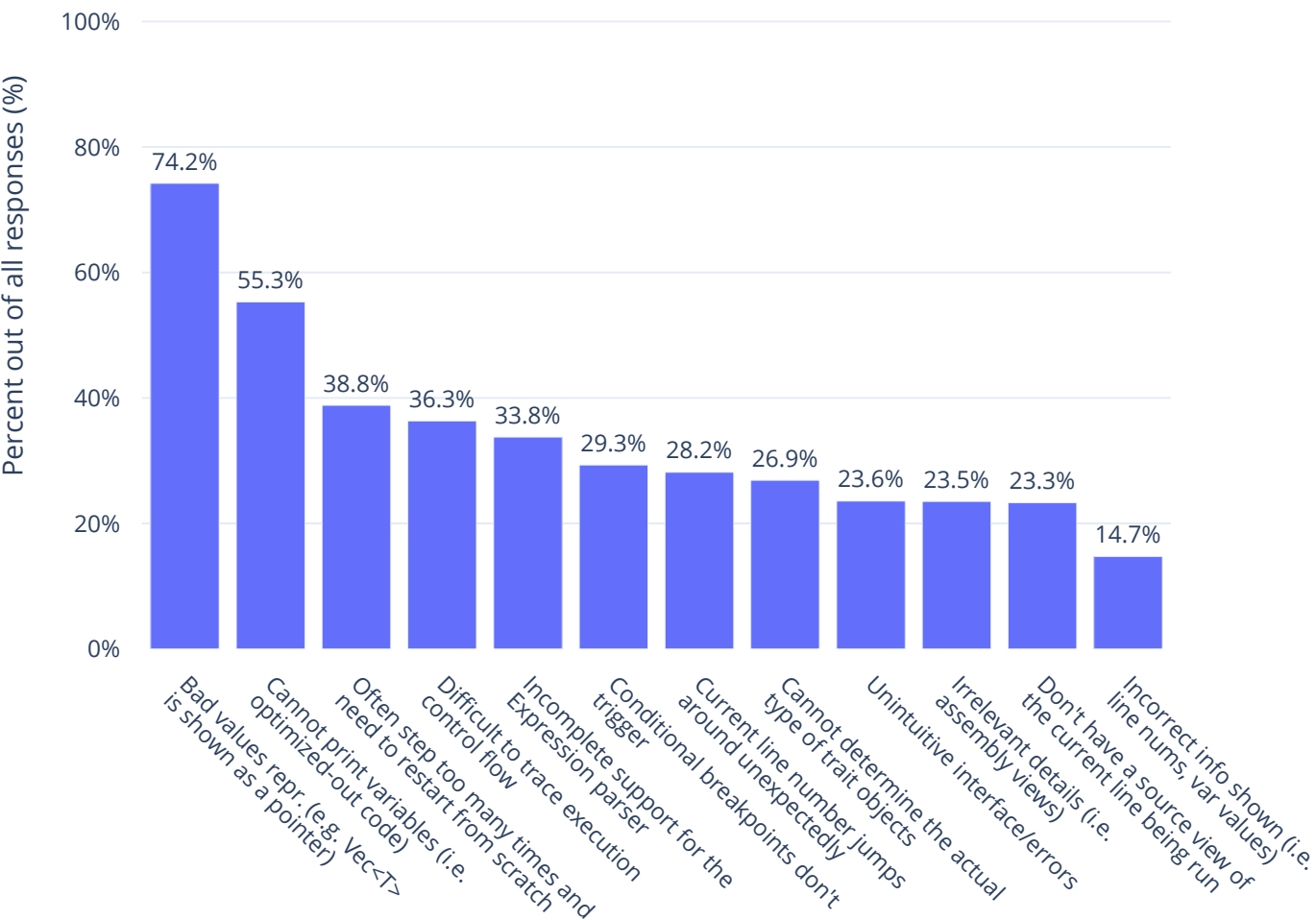


Wordcloud of open answers for the previous chart:



Which of these pain points have you experienced using a debugger with Rust?

(total responses = 1481, multiple answers)



Wordcloud of open answers for the previous chart:

